University of Glasgow

Level 0

OKD4 Cluster

Understanding your account, resources and storage

Dr. Richard McCreadie

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How Do I Use it?

What you need to do will define what you need to learn to use the cluster, you can think of it as having 4 levels, where you incrementally need to understand more of how the cluster works as you need to undertake more complex work

You are Here

Level 0: Understanding your account, resources and storage

Level 1

- Cluster users who just need to get up and running with Python/Jupyter or need remote execution with VSCode are recommended to use the Launcher tool

Level 2

- For scenarios where you need to deploy a wider range of existing services, you will need to learn the basics of Containers and Kubernetes. This will allow you to deploy pre-built container solutions from platforms like Dockerhub or more complex services via OperatorHub.

Level 3

- For running batch experiments (queuing many containers to run concurrently) you will need to learn how to interact with the cluster via the command line tool (oc)

Level 4

- If you need to create a new development environment from first principles or custom applications/demos, then you will need to learn how to produce new containers via Builds, and it is recommended to learn how to create continuous integration pipelines.

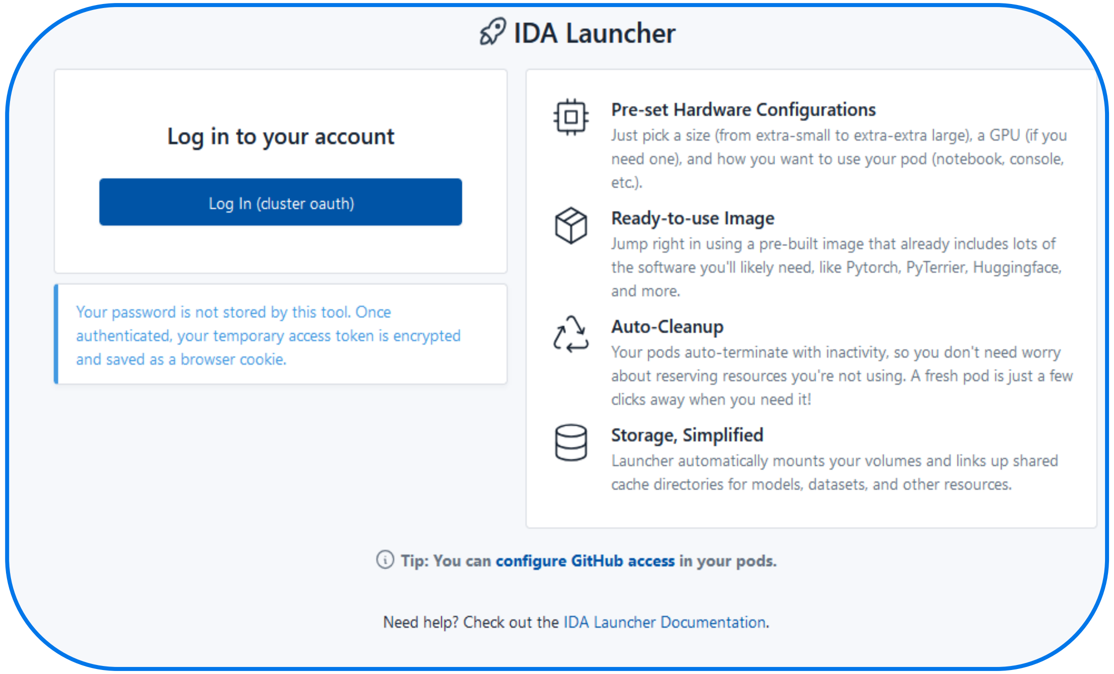
Components of your OKD4 Account

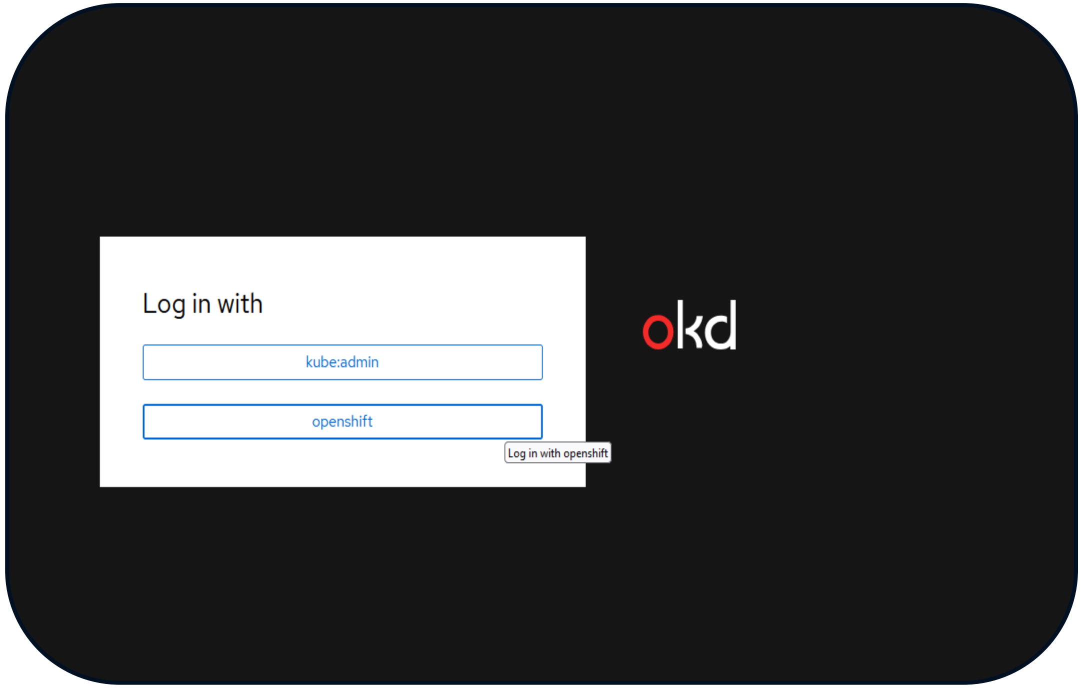
To start your OKD4 journey we need to introduce some basic concepts such that you understand what you have available to you, your OKD4 account is comprised of:

- Unique Identifier: This is typically of the form <accountType><creationYear><firstname>
    - E.g. for a PhD student: pgr24Richard
- Password: By default this will have been generated by the cluster administrator
- Project: You can think of this as a sandbox that you own and can deploy software into, and shares its name with your identifier
- Personal Storage: You are given a folder into which you can persistently store data on the remote storage server(s), more on this later.
- Shared Storage: You are given access to a set of shared directories containing datasets, LLMs, indices and other resources.
- Resource Quota: Depending on the account type you will be assigned a maximum amount of resources you can request at any one time

Logging-in

- Most users interact with the cluster via web browser, whether it be using Launcher or the Cluster Console, you will login to these using your identifier and password

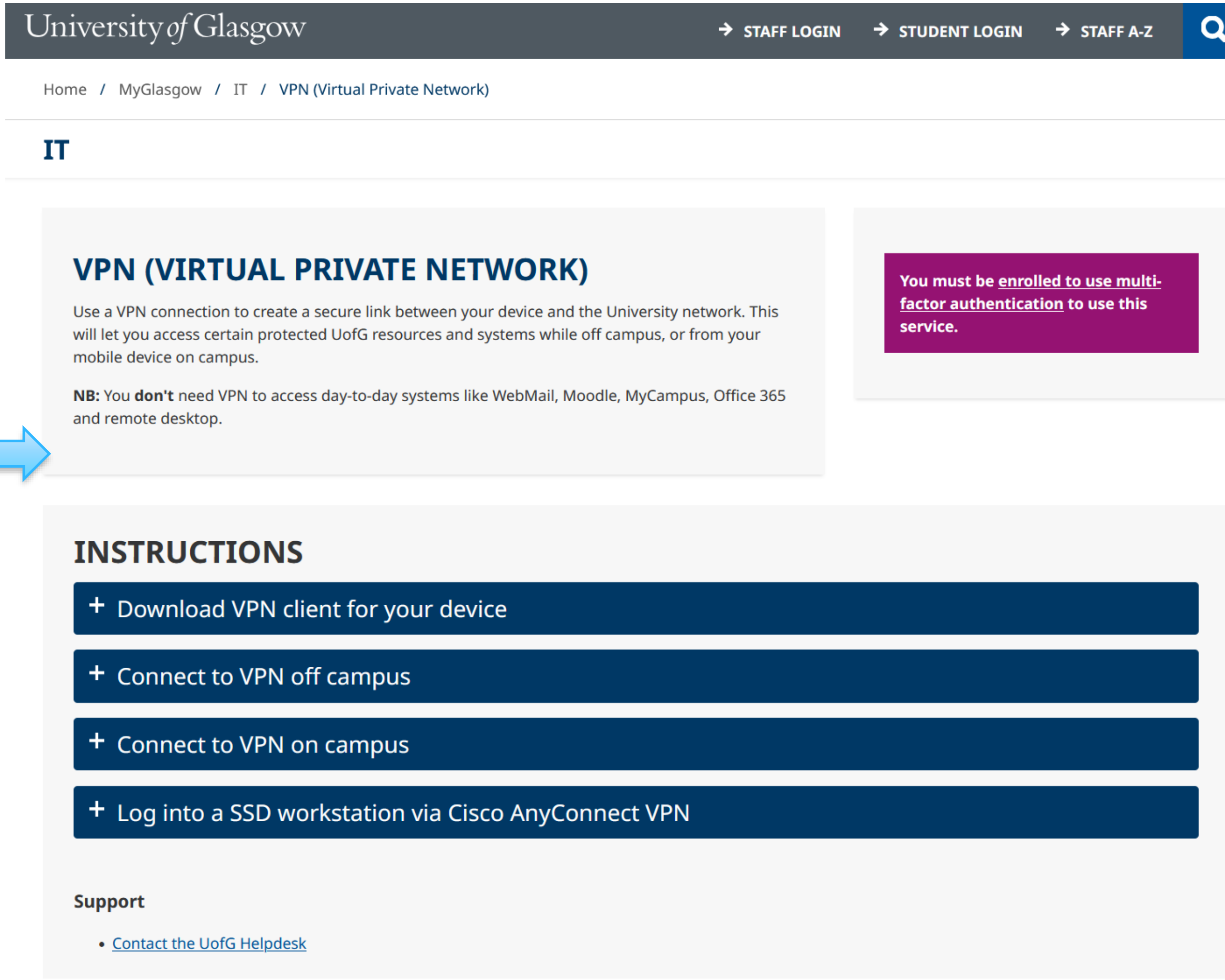




...under the hood this assigns you an OAuth token that grants access for a while

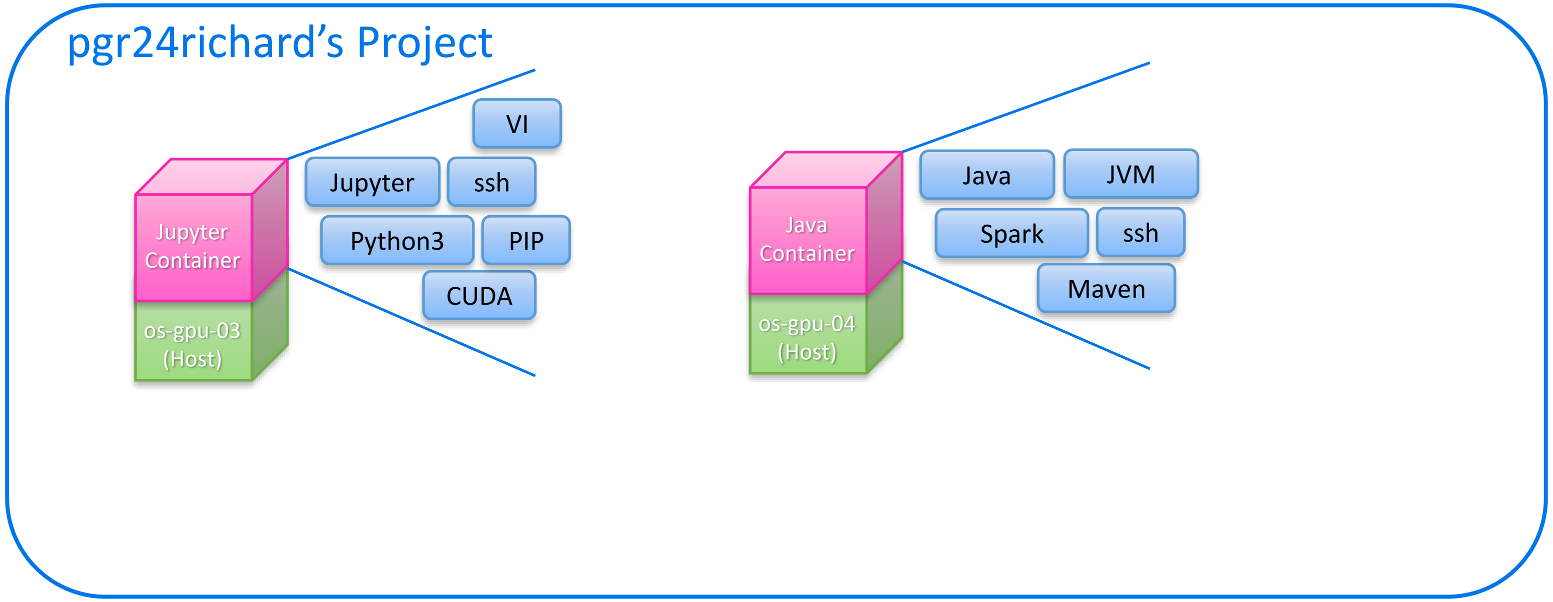
Network Access

- The cluster is physically located within the Computing Science building, for security those machines are not visible to the outside world
- Therefore, to access the cluster you need to be either
  - Connected to the University (Cisco) VPN
    - https://www.gla.ac.uk/myglasgow/it/vpn/
  - Connected to the School VPN
  - Or physically connected (via ethernet) to the School network with a white-listed device (like a university provided desktop computer)



Your Project and Containers

- Your project is a virtual sandbox where your software will run
  - More precisely, it is a Namespace where Containers will be Executed
- A Container is a pre-built bundle of code, libraries and configuration, which in combination with a host machine defines an execution environment
  - You can think of it like a blueprint for Unix installation, that can be replicated on the cluster machines
  - There are multiple types of container, but the most common are Docker containers



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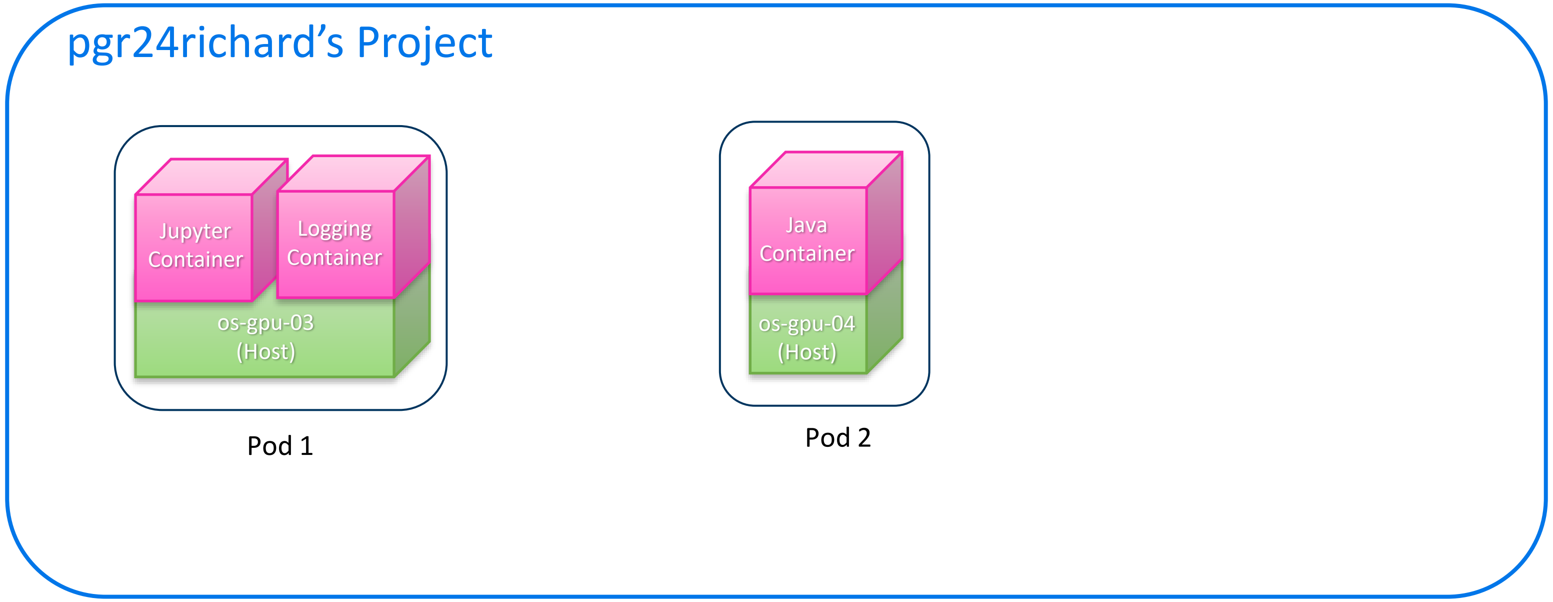
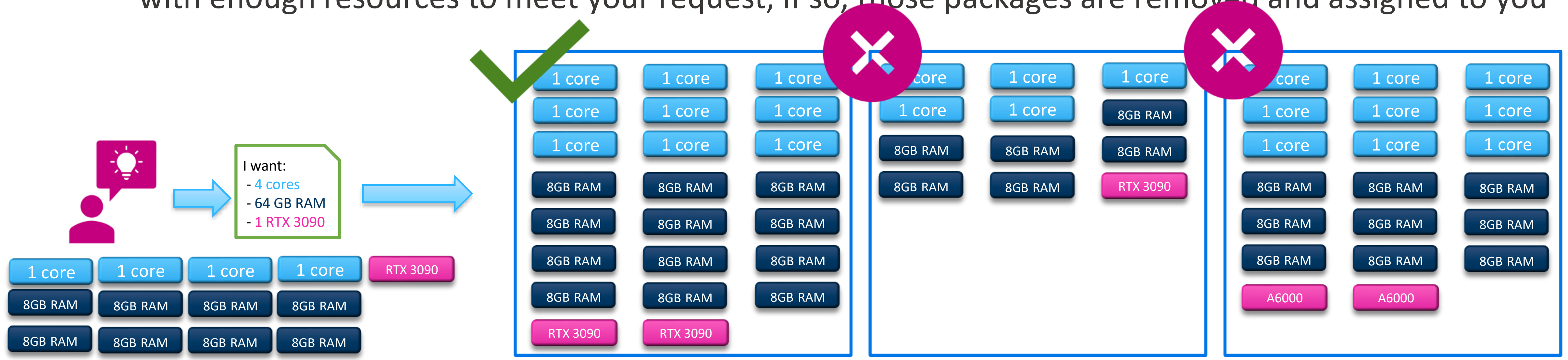
  - A Pod is a grouping of multiple containers with the same host



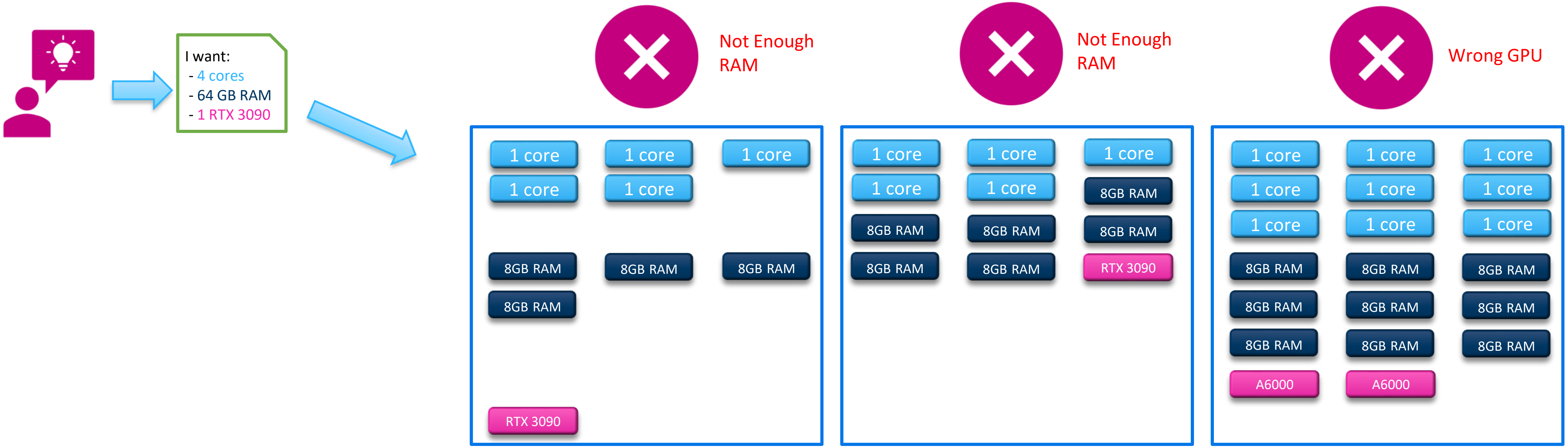
Understanding Compute Resources

- When we want the cluster to run a container, we are going to need to ask it for some compute resources to do that work
- Conceptionally, you can think of the cluster as a series of warehouses (representing physical machines/hosts) containing three types of pre-packed boxes
  - Those containing some CPU cores
  - Those containing RAM
  - Those containing a GPU
- When you ask for resources, a scheduler looks to see if there are any of the warehouses/machines with enough resources to meet your request, if so, those packages are removed and assigned to you



Understanding Compute Resources

- If we ever ask for a set of resources and the cluster cannot provide them then it will report that your workload is Unschedulable (because the scheduler failed to find the resources)
  - Note that the scheduler will regularly keep re-checking (every few seconds)
- You may also have realised that it is possible to exhaust one type of resource before another, i.e. a machine might have all its RAM allocated, but still have a GPU free
  - This means that it is important to only request the resources that you actually need



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- As I mentioned earlier, your OKD account also has a resource quota
  - This tells the scheduler the maximum resources you can hold at any one time

Persistent Storage

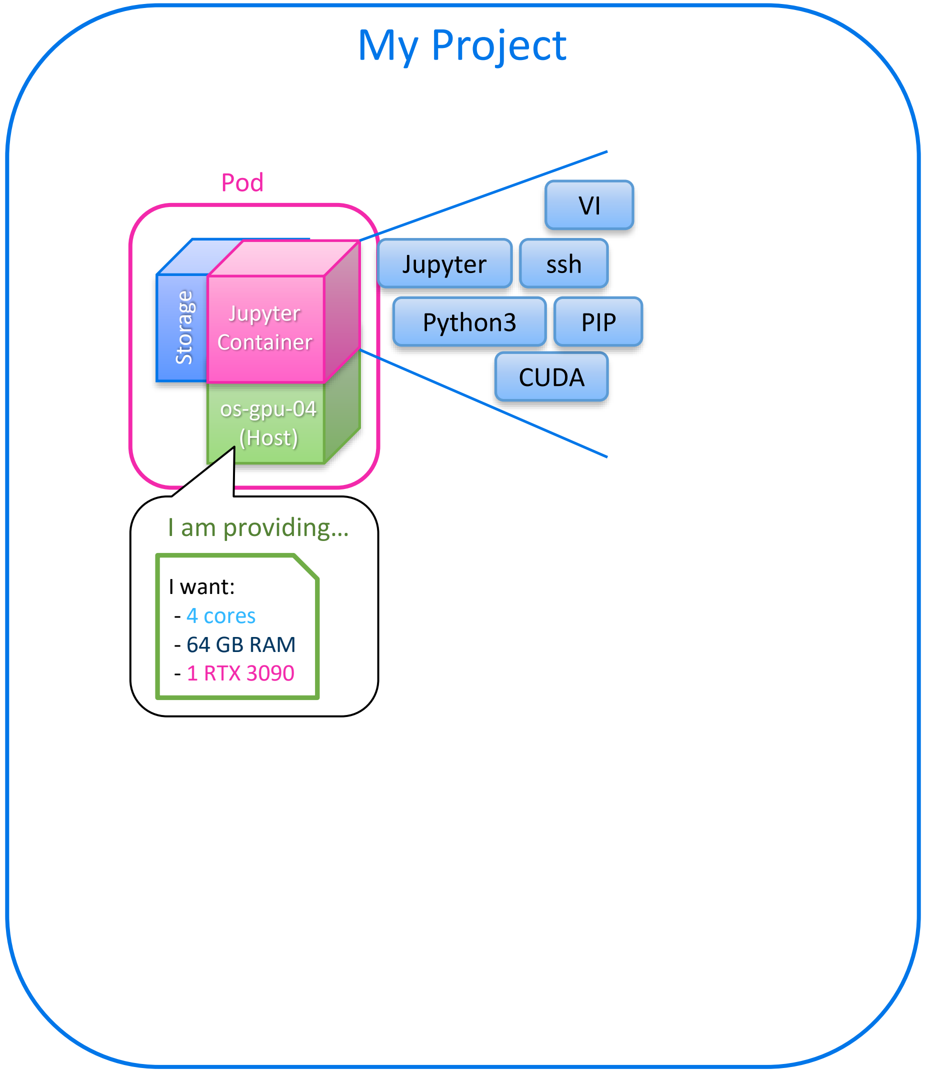
- All the work that we do requires us to store data, e.g.
  - An input corpus
  - An LLM model
  - The output of an experiment

- Any information stored within a container is deleted when that container exits
  - So, we need a way to persist data between container executions

- This is the role of persistent storage
  - You can attach your personal storage folder to any (and all) containers that you run
  - Data written into that folder will persist
  - You can also attach shared directories to your containers if you need the data that they hold

Basic Cluster Mental Model

- To summarize
  - You have a sandbox called a Project
  - Inside that project we can run Containers
    - … which hold our code and libraries
  - When we want to run a container we need to ask for resources
    - A scheduler will look at the available machines/hosts in the cluster to find one with enough resources, and then assign the container to that host
  - Finally, we attach persistent storage to the container such that we have somewhere to read from and write to
  - A Pod is a set of one or more containers on the same host



Questions?

Dr. Richard McCreadie

Real-time IR, Machine Learning, Big Data Stream Processing, Evaluation

Richard.McCreadie@glasgow.ac.uk